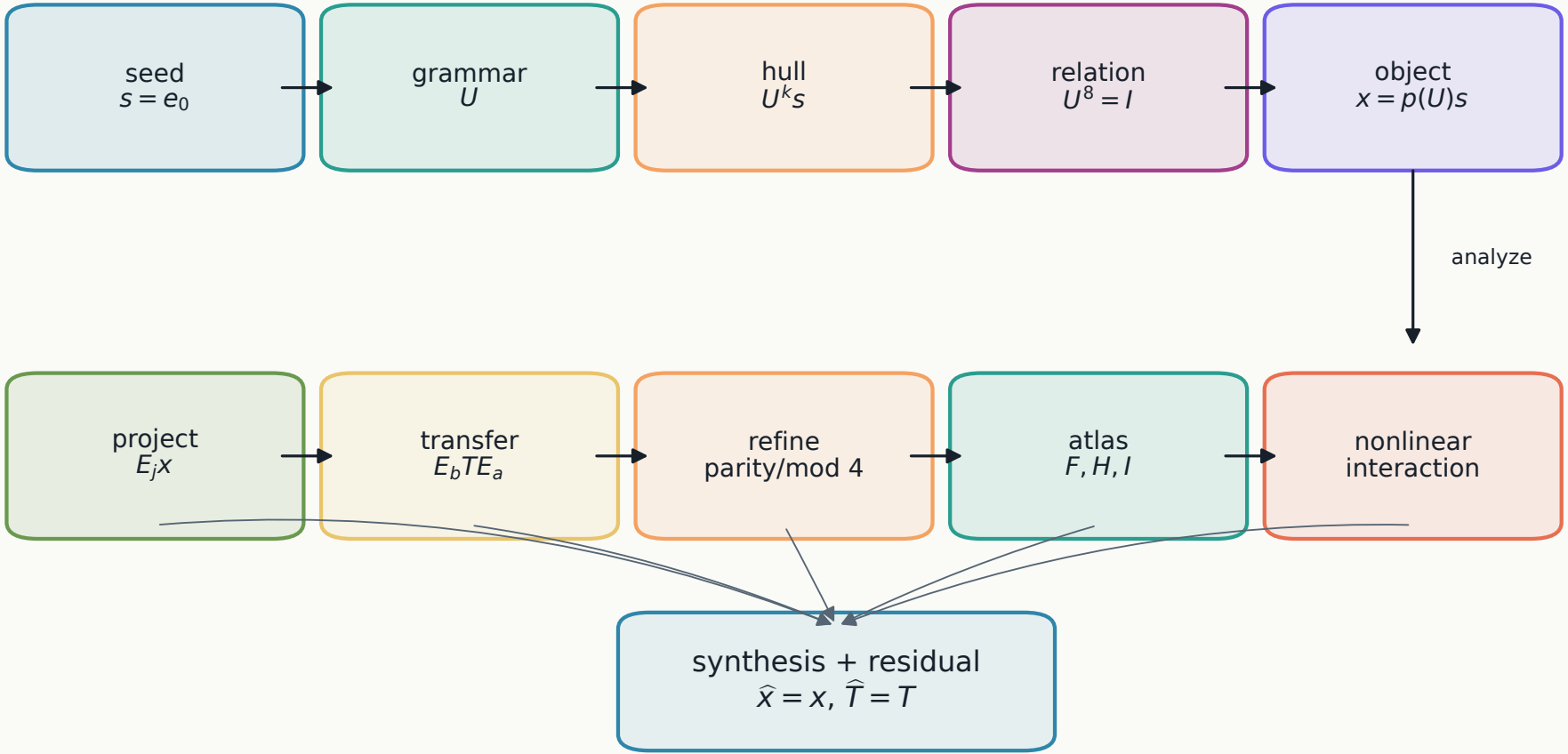Master example: the length-eight cyclic signal through the complete calculus